

Chapter 7 7.1 David Goodsell; **p. 129 mosaic** camerawithlegs/Fotolia; **7.10** Used by permission of B. L. de Groot. Related to work done for: Water Permeation Across Biological Membranes: Mechanism and Dynamics of Aquaporin-1 and GlpF. B. L. de Groot, H. Grubmüller. *Science* 294:2353-2357 (2001); **p. 132 Agre** Sam Kittner, Pearson Education; **7.14 Michael Abbey**/Science Source; **p. 135 ion channel** From: Crystal structure of a mammalian voltage-dependent Shaker family K⁺ channel. S. B. Long et al. *Science*. 2005 Aug 5;309(5736):897-903. Epub 2005 Jul 7. Cover image; **p. 135 Serrano** Darren Phillips/New Mexico State University; **Scientific Skills Exercise** Photo Fun/Shutterstock; **7.21 left** Biophoto Associates/Science Source; **center** Don W. Fawcett/Science Source; **right** From: M.M. Perry and A.B. Gilbert, *Journal of Cell Science* 39: 257-272, Figs. 11 and 13 (1979). © 1979 The Company of Biologists Ltd.; **p. 142 spray** Kristoffer Trippelaar/Alamy Stock Photo.

Chapter 8 8.1 Biophoto/Alamy Stock Photo; **8.2** Stephen Simpson/Getty Images; **8.3** Robert N. Johnson/RnJ Photography; **8.4a** From: Micromechanical properties of biological silica in skeletons of deep-sea sponges. Alexander Woesz et al. *J. Mat. Res.* Volume 21, Issue 8, August 2006, pp. 2068-2078. Fig. 1; **8.4b** Neale Clark Agency/robertsharding/Alamy Stock Photo; **8.15 Thomas Steitz**; **Scientific Skills Exercise** Fer Gregory/Shutterstock; **8.17** Jack Dykinga/Nature Picture Library/Alamy Stock Photo; **8.22** Keith R. Porter/Science Source; **p. 163 penguins** Flickr/Getty Images.

Chapter 9 9.1 Tierfotoagentur/Alamy Stock Photo; **9.3** Dionisvera/Fotolia; **9.9** Z. Hong Zhou, University of California, Los Angeles; **Scientific Skills Exercise** Thomas Kitchin & Victoria Hurst/Design Pics/Alamy Stock Photo; **p. 180 cacao** Aedka Studio/Shutterstock; **p. 186 model** Medical Research Council; **CoQ10** Stephen Rees/Shutterstock.

Chapter 10 10.1 tree Rolf Roeckl/mauritius images GmbH/Alamy Stock Photo; **caterpillar** blickwinkel/Alamy Stock Photo; **10.2a** STILLFX/Shutterstock; **10.2b** Natalie Jean/Shutterstock; **10.2c** M. I. Walker/Science Source; **10.2d** Michael Abbey/Science Source; **10.2e** Heide N. Schulz-Vogt, Leibniz Institute for Baltic Sea Research Warnemuende; **10.3 top**, **10.21** Andreas Holzenburg, University of Texas Rio Grande Valley; **bottom** Jeremy Burgess/Science Source; **p. 194 C. thermalis** Dennis Nürnberg; **10.11b** Christine L. Case; **Scientific Skills Exercise** The Ohio State University; **10.20a** Doukdouk/Alamy Stock Photo; **10.20b** Keysurfing/Shutterstock; **p. 211 snow** gary yim/Shutterstock.

Chapter 11 11.1 top, 11.5c Federico Veronesi/Gallo Images/Alamy Stock Photo; **bottom** Nature Picture Library/Alamy Stock Photo; **11.2a-c** A. Dale Kaiser/Stanford University; **11.2d** Michiel Vos; **Problem-Solving Exercise** Bruno Coignard and Jeff Hageman, CDC; **p. 214 Bassler** Alena Soboleva; **11.7** From: High-resolution crystal structure of an engineered human beta2-adrenergic G protein-coupled receptor. V. Cherezov, et al. *Science*. 2007 Nov 23;318(5854):1258-65. Epub 2007 Oct 25; **p. 220 Bautista** Mark Joseph Hanson; **11.19** Gopal Murti/Science Source; **11.21** William Wood; **p. 233 chips** Maureen Spuhler/Seelevel.com.

Chapter 12 12.1 George von Dassow; **12.2a, c** Biophoto/Science Source; **12.2b** Biology Pics/Science Source; **12.3** Andrew S. Bajer, University of Oregon, Eugene; **12.4**, **12.5** Biophoto/Science Source; **12.7** Conly L. Rieder, Wadsworth Center, Albany, NY; **12.8 left** Jane Stout and Claire Walczak, Indiana University; **right** Matthew J. Schibler; **12.10a** Don W. Fawcett/Science Source; **12.10b** A. Palevitz and E. H. Newcomb, University of Wisconsin; **12.11** Elizabeth Pierson, Pearson Education; **p. 246 Nurse** Zach Veilleux, Rockefeller University; **12.18** Guenter Albrecht-Buehler; **12.19** Ian Bo Chen; **12.20** Nature's Geometry/Science Source; **Scientific Skills Exercise** Molecular Expressions; **p. 250 Alberts** Photo by Tom Kochel, courtesy of the Department of Biochemistry and Biophysics, The University of California San Francisco. Gitschier J. (2012) "Scientist Citizen: An Interview with Bruce Alberts". *PLoS Genetics* 8(5): e1002743. doi:10.1371/journal.pgen.1002743. Figure 1; **p. 252 onion cells** Scenics & Science/Alamy Stock Photo; **HeLa cells** Steve Gschmeissner/Science Source.

Unit Three Interview Mojica, flask, petri dish University of Alicante; lake Richard Brown/Alamy Stock Photo.

Chapter 13 13.1 PeopleImages/DigitalVision/Getty Images; **p. 255 sperm** Don W. Fawcett/Science Source; **13.2a** Roland Birke/Okapia/Science Source; **13.2b** George Ostertag/Alamy Stock Photo; **13.3 top** Ermakoff/Science Source; **bottom** CNRI/Science Source; **Scientific Skills Exercise** SCMAT/Science Source; **13.12** Mark Petronczki and Maria Siomos; **13.13** John Walsh, Micrographia.com; **p. 268 bananas** Randy Ploetz.

Chapter 14 14.1 John Swithinbank/Agfotostock; **14.14a** Maximilian Weinzierl/Alamy Stock Photo; **14.14b** Paul Dymond/Alamy Stock Photo; **Scientific Skills Exercise** Apomares/E+/Getty Images; **14.15** Barbara Bowman, Pearson Education; **14.16** Patricia Willocq; **14.18** Michael Ciesielski Photography; **p. 287 Wexler** Ron Galella, Ltd./Getty Images; **14.19** CNRI/Science Source; **p. 293 cat** Arco Images GmbH/Alamy Stock Photo; **family** Rene MALTETE/Gamma-Rapho/Getty Images.

Chapter 15 15.1 Courtesy of Peter Lichter; **15.2** Martin Shields/Alamy Stock Photo; **15.5** Andrew Syred/Science Source; **15.6b** Li Jingwang/E+/Getty Images; **15.6c** Kosam/Shutterstock; **15.6d** Creative images/Fotolia; **15.8** Jagodka/Shutterstock; **Scientific Skills Exercise** Oliver911119/Shutterstock; **p. 306 Orr-Weaver** Maria Nemchuk; **15.15 left** CNRI/Science Source; **right** Denys_Kuvaiev/Fotolia; **p. 311 Tilghman** Denise Applewhite, Office of Communications, Princeton University; **15.18** Phomphan/Shutterstock; **p. 313 butterfly** James K Adams.

Chapter 16 16.1 4X-image/E+/Getty Images; **16.3** Oliver Meckes/Eye of Science/Science Source; **Scientific Skills Exercise** Marevision Agency/AGE Fotostock/Alamy Stock Photo; **16.6a** Library of Congress; **16.6b** Science Source; **16.13a** Jerome Vinograd; **16.13b** From: Enrichment and visualization of small replication units from cultured mammalian cells. D. J. Burks et al. *J Cell Biol.* 1978 Jun;77(3):762-73. Fig. 6A; **16.22** Peter Lansdorp; **16.23 DNA strand** Gopal Murti/Science Source; **nucleosome** Victoria E. Foe; **chromosome** Biophoto/Science Source; **16.24a** Thomas Reid, Genetics Branch/CCR/NCI/NIH; **16.24b** Michael R. Speicher/Medical University of Graz; **p. 334 models** Thomas A. Steitz/Yale University.

Chapter 17 17.1 AFP/Getty Images; **17.7a** Keith V. Wood; **17.7b** Sinclair Stammers/Science Source; **p. 346 Steitz** National Science Foundation; **17.18** Joachim Frank; **17.23b** Barbara Hamkalo; **17.24** Oscar Miller/Science Source; **17.26** Eye of Science/

Science Source; **Problem-Solving Exercise** Duplass/Shutterstock; **p. 360 Mojica** University of Alicante; **p. 364 cat** Vasily Koval/Shutterstock.

Chapter 18 18.1 top gallimaufry/Shutterstock; **18.1 bottom** Andreas Werth; **18.8a** Permission for use granted by Randy Jirtle, Professor of Epigenetics, NC State University, Raleigh, NC; **18.8b** History and Art Collection/Alamy Stock Photo; **Scientific Skills Exercise** hideos/E+/Getty Images; **18.13** Michael Speicher and Nigel Carter, Medical University of Graz; **p. 381 Lin** Courtesy of Haifan Lin; **18.16** Mike Wu; **18.20** F. Rudolf Turner, Indiana University; **18.21** Wolfgang Driever, University of Freiburg, Freiburg, Germany; **p. 387 Hopkins** Courtesy of the Cold Spring Harbor Laboratory Archives; **18.22** Ruth Lahmann, The Whitehead Institution; **18.27** Bloomberg/Getty Images; **p. 394 King** © University of Washington; **p. 397 fish** Peter Herring/Image Quest Marine.

Chapter 19 19.1 Thomas Deerinck, NCMIR/Science Source; **19.2** Peter von Sengbusch, Botanik; **19.3a** Science Source; **19.3b** Linda M. Stannard, University of Cape Town/Science Source; **19.3c** Hazel Appleton, Health Protection Agency Centre for Infections/Science Source; **19.3d** Ami Images/Science Source; **19.7** molekuul.be/Fotolia; **p. 404 Mojica** University of Alicante; **19.9 top** Charles Dauguet/Science Source; **bottom** Petit Format/Science Source; **19.10a** CDC; **19.10b** Kuhn and Rossmann research groups, Purdue University; **19.10c** Cynthia Goldsmith, CDC; **p. 410 Satcher** Courtesy of David Satcher; **Scientific Skills Exercise** Dong yanjun/Imaginechina/AP Images; **19.11** Olivier Asselin/Alamy Stock Photo; **inset** James Gathany, CDC; **19.12** Nigel Cattlin/Alamy Stock Photo; **p. 414 Tamiflu** Nelson Hale/Shutterstock.

Chapter 20 20.1 Ian Derrington; **20.2** P. Morris, Garvan Institute of Medical Research; **20.6b** Scott Sinkler/Alamy Stock Photo; **20.9** Ethan Bier; **20.13** George S. Watts and Bernard W. Futscher, University of Arizona Cancer Center; **20.14** Stephen McNally, UC Berkeley; **p. 427 Rotimi** National Human Genome Research Institute; **20.17** Roslin Institute; **20.18** Pat Sullivan/AP Images; **20.19** Steve Gschmeissner/Science Photo Library/Alamy Stock Photo; **20.23** Brad DeCecco/Redux; **20.24** Steve Helber/AP Images; **p. 439 Suzuki** dpa picture alliance archive/Alamy Stock Photo; **p. 441 hot spring** Galyna Andrushko/Shutterstock.

Chapter 21 21.1 elephant shark Image Quest Marine; sea horse Rich Cary/Shutterstock; **21.4** University of Toronto Lab; **21.5** Affymetrix; **21.7 left** AP Images; **right** Virginia Walbot; **p. 451 Walbot** Courtesy of Virginia Walbot; **21.10a** Oscar L. Miller Jr., Dept. of Biology, University of Virginia; **p. 455 Lander** Justin Knight Photography; **21.18** mept Francois Gohier/Science Source; **brain cells** From: Altered ultrasonic vocalization in mice with a disruption in the *Foxp2* gene. W. Shu et al. *Proc Natl Acad Sci U S A*. 2005 Jul 5;102(27):9643-8. Epub 2005 Jun 27. Fig. 3; **p. 463 Rotimi** National Human Genome Research Institute; **p. 464 sea urchins** WaterFrame/Alamy Stock Photo; **p. 466 treehopper** Patrick Landmann/Science Source.

Unit Four Interview top Kris Snibbe; bottom Courtesy of Cassandra Extavour.

Chapter 22 22.1 Lighthouse/UiG/AGE Fotostock; **22.2 rhino** *Recherches sur les ossements fossiles*. G. Cuvier. Atlas, pl. 17 (1836); **iguana** Wayne Lynch/All Canada Photos/AGE Fotostock; **frog** The Natural History Museum, London/Alamy Stock Photo; **Alfred Russel Wallace** The Natural History Museum/Alamy Stock Photo; **The Origin of Species** 1859 Murray edition of *The Origin of Species* by Charles Darwin; **22.4** Karen Moskowitz/Stone/Getty Images; **22.5 left** Derek Bayes/Lebrecht Music and Arts Photo Library/Alamy Stock Photo; **right** Photo Researchers/Science History Images/Alamy Stock Photo; **22.6a** Michel Gunther/Science Source; **22.6b** David Hosking/Frank Lane Picture Agency; **22.6c** David Hosking/Alamy Stock Photo; **22.7** Darwin, C. R. Notebook B: Transmutation of species (1837-1838), p. 36. CUL-DAR121; **22.9 Brussels sprouts** Arena Photo UK/Fotolia; **kale** Željko Radojko/Fotolia; **cabbage** Guy Shapira/Shutterstock; **wild mustard** Martin Fowler/Alamy Stock Photo; **broccoli** YinYang/E+/Getty Images; **kohlrabi** Motorolka/Shutterstock; **22.10** Robert Hamilton/Alamy Stock Photo; **22.11** Kichigin/Shutterstock; **22.12a** William Mullins/Alamy Stock Photo; **22.12b** Chris Mattison/Alamy Stock Photo; **22.13** Scott P. Carroll; **22.16 left** Keith Wheeler/Science Source; **right** Omikron/Science Source; **22.18 left** ANT Photo Library/Science Source; **right** Joe McDonald/Steve Bloom Images/Alamy Stock Photo; **22.19** Chris Linz, Thewissen Lab, Northeastern Ohio Universities College of Medicine (NEOUCOM); **p. 485 honeypot ant** © Joel Sartore/National Geographic Image Collection.

Chapter 23 23.1 Sylvain Cordier/Science Source; **p. 487 Grants** Robin Heyden; **23.3** Juniors Bildarchiv GmbH/Alamy Stock Photo; **23.5** Erick Greene; **23.6 left** Edward Bennett Agency/Design Pics Inc/Alamy Stock Photo; **right** Patrick Valkenburg/Alaska Department of Fish and Game; **Scientific Skills Exercise** DLeonis/Fotolia; **23.11** Bruce Montagne/Dembinsky Photo Associates/Alamy Stock Photo; **23.12 top** Kristin Stanford, Stone Laboratory, Ohio State University; **bottom** Kent Bekker, United States Fish and Wildlife Service; **23.14** Anthony Bannister/NHPA/Photoshot/Newscom; **23.15** Dave Blackey Agency/All Canada Photos/Alamy Stock Photo; **23.18 blood cells** Eye of Science/Science Source; **doctor with child** Caroline Penn/Alamy Stock Photo; **mosquito** James Gathany and Frank Collins, University of Notre Dame, CDC; **23.19a** From: Camouflage mismatch in seasonal coat color due to decreased snow duration. L. S. Mills, et al. *Proc Natl Acad Sci U S A*. 2013 Apr 30;110(18):7360-5. Fig. 1; **23.19b** L. Scott Mills; **p. 505 lake** Thomas/Pat Leeson/Science Source.

Chapter 24 24.1 Joel Sartore/National Geographic Image Collection; **24.2a left** Malcolm Schuyf/Alamy Stock Photo; **right** Wave RF/Getty Images; **24.2b top left** Robert Kneschke/Kalium/AGE Fotostock; **top center** Justin Horrocks/E+/Getty Images; **top right** Ryan Mcvay/Getty Images; **bottom left** Dragon Images/Shutterstock; **bottom center** arek_malang/Shutterstock; **bottom right** Jaki good photography - celebrating the art of life/Moment Open/Getty Images; **24.3a** Phil Huntley-Franck; **24.3b** Jerry A. Payne, USDA Agricultural Research Service, Bugwood.org; **24.3c** Hogle Zoo; **24.3d** USDA; **24.3e** Imagebroker/Alamy Stock Photo; **24.3f, g** Takahiro Asami; **24.3h** Larry Geddis/Alamy Stock Photo; **24.3i** Chuck Brown/Science Source; **24.3j** mivod/Shutterstock; **24.3k** Bagicat/Fotolia; **24.3l** FreeReinDesigns/Fotolia; **24.3m** Kazutoshi Okuno; **24.4 top** CLFProductions/Shutterstock; **right** Boris Karpinski/Alamy Stock Photo; **bottom** Troy Maben/AP Images; **24.6a** Courtesy of Brian Langerhans; **24.8 maps** NASA EOS Earth Observing System; **shrimp** Arthur Anker, Florida Museum of Natural History; **Scientific Skills Exercise** John Shaw/Avalon/Photoshot/Alamy Stock Photo; **24.11** Pam and Doug Soltis; **24.12** Ole Seehausen; **24.13** Jeroen Speybroeck, Research